International Islamic University Chittagong

Department of Computer Science and Engineering
B.Sc. in CSE, Semester Final Examination

Session: Spring-2025
Course Code: CSE-3631
Time: 2 hour 30 Minutes

Semester: 6th
Course Title: Operating System
Marks: 50

(i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes is mentioned in additional Columns

Course Learning Outcomes (CLOs):	
CLO1	Explain the fundamental principles and core services provided by an operating system
CLO2	Analyze and evaluate scheduling algorithms for process management and synchronization mechanisms to ensure efficient resource allocation.
CLO3	Evaluate and apply memory management techniques, virtual memory concepts, file system structures and disk management techniques to optimize resource utilization.

Group A

			CLO	Marks
Q1		A system has 4 processes and 5 allocate resource. The current allocation and maximum needs are as follows-	3	10

	Allocation					Maximum				
	A	B	C	D	E	A	B	C	D	E
P1	1	0	2	1	1	1	1	2	1	3
P2	2	0	1	1	0	2	2	2	1	0
P3	1	1	0	1	1	2	1	3	1	1
P4	1	1	1	1	0	1	1	2	2	0

Using Banker's algorithm, answer the following questions: -

i) What are the contents of need matrix?

ii) If initial value = [0 0 X 1 1], what is the smallest value of x for which this is a safe state? find the safe sequence.

iii)If a request from process P3 arrives for (0 0 1 1 1), can the request be granted immediately? If yes find the safe stat.

Or

Q1	a)	A virtual memory system employs a main memory with 4-page frames. Consider the following page reference string generated by the CPU: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3. Demonstrate the application of the Optimal Page Replacement Algorithm for the given page reference string and frame size. For each page reference, clearly indicate whether a page fault (F) or a page hit (H) occurs. Calculate and state the total number of page faults incurred. Justify your eviction choices at each step where a page replacement is necessary by referring to the future page references.	3	5
	b)	A foundational concept in operating systems is efficient memory management, particularly when physical memory is limited. Assume a system is configured with a main memory comprising 3-page frames. The CPU generates the following page reference string: 1, 3, 0, 3, 5, 6, 3. Applying the First-In, First-Out (FIFO) Page Replacement Algorithm, systematically trace the page replacements for the given reference string and frame configuration. For each	3	5

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page reference processed, explicitly denote whether it results in a page fault (F) or a page hit (H). Calculate and present the total number of page faults observed throughout the entire sequence.

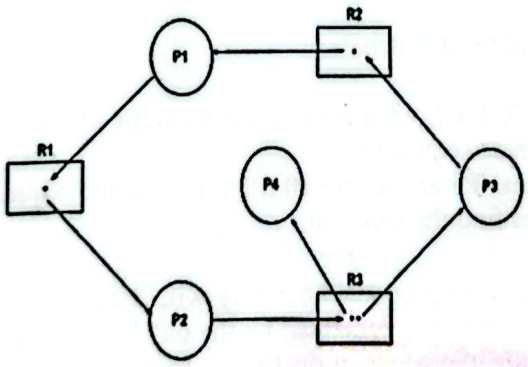
Q2	a)	Deadlock represents a critical state within a multi-programmed operating system where a set of processes becomes permanently blocked, each holding a resource and simultaneously waiting for a resource held by another process in the set. Analyze and articulate the necessary conditions that must simultaneously hold true for a deadlock to arise in a system. For each condition, explain its individual contribution to the deadlock scenario.	2	3
	b)	What are the key factors that influence the performance of demand paging? Explain the concept of the Effective Memory Access Time (EMAT) in a paged system. How is it calculated?	2	4
	c)	Discuss the potential drawbacks and challenges associated with deadlock recovery	3	3

Group B

Q3	a)	Consider a swapping system in which memory consists of the following hole sizes in memory order: 15KB, 8KB, 113KB, 10KB, 7KB and 18KB. Which hole is taken for successive segment requests (Representation by diagram is preferable): i. 113 KB ii. 115 KB iii. 7 KB iv. 20 KB v. 7 KB for the first fit, best fit and worst fit? Which makes the most efficient use of memory?	3	4
	b)	In spite of limited physical memory we can execute of program whose logical memory size is greater than physical memory of the system. How it's possible?	2	3
	c)	Describe steps in handling a page fault with proper diagram.	2	3

Or

Q3



	a)	Consider the above-mentioned multiple instances resource allocation graph for determining if the state is safe or not and also provide an allocation and request matrix to justify the outcome.	2	5
	b)	Deadlock represents a critical state within a multi-programmed operating system where a set of processes becomes permanently blocked, each holding a resource and simultaneously waiting for a resource held by another process in the set. Analyze and articulate the necessary conditions that must simultaneously hold true for a deadlock to arise in a system. For each condition, explain its individual contribution to the deadlock scenario.	2	5
Q4	a)	A virtual memory system employs a main memory with 4 page frames. Consider the following page reference string generated by the CPU: a, b, c, d, e, e, b, f, c, b, c, d, b, a, b, c. Demonstrate the application of the Optimal Page Replacement	2	5

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Algorithm for the given page reference string and frame size. For each page reference, clearly indicate whether a page fault (F) or a page hit (H) occurs. Calculate and state the total number of page faults incurred. Justify your eviction choices at each step where a page replacement is necessary by referring to the future page references.

b) Virtual memory, a cornerstone of modern operating systems, fundamentally abstracts the concept of physical memory, allowing programs to operate as if they have access to a very large address space. This abstraction is commonly realized through demand paging, a technique where processes reside in secondary memory, and pages are loaded into main memory only when they are actively referenced, rather than pre-loaded in their entirety. Elaborate on the core mechanism of a demand paging system. Explain how it leverages the principles of virtual memory and paging with swapping to enable execution of programs larger than physical memory. **2** **5**

Q5 a) Why does thrashing occurs? Explain the working set model in thrashing. **2** **3**

b) What is file allocation method? Describe contiguous and Linked allocation method. **2** **2**

c) Consider an access matrix for a system with three subjects (S1, S2, S3) and four objects (O1, O2, O3, O4). Each entry in the matrix represents the access rights for a subject to an object, where 0 means no access, 1 means read access, and 2 means write access. The access matrix is as follows: **3** **5**

	O1	O2	O3	O4
S1	1	2	0	1
S2	0	1	1	0
S3	2	0	0	1

Calculate the total number of access rights granted in the access matrix.

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